SOFTWARE PROJECT PROPOSAL

AI-Powered LMS, Advertising Toolkit & Gamified Community Platform

Inspired by AdLevel.ai · Skool · Whop

Prepared By: Senior Full-Stack & AI Engineer
Billing Model: Hourly — Weekly Accumulated Hours
Rate: $16 USD / Hour
Estimated Project Investment: $12,800 – $14,400 USD
Estimated Timeline: 26 Weeks to Full Production Launch

CONFIDENTIAL — Prepared for Client Review

1. Executive Summary

1.1 What We Are Building

This proposal covers the complete development of a premium SaaS platform that unifies three high-value product categories under a single subscription: an AI-powered marketing and advertising toolkit, a structured video-based learning management system (LMS), and a gamified community portal. The platform is further differentiated by multi-tenant architecture, enabling white-labeling and future marketplace expansion. Anthropic's Claude API powers the intelligence layer across every major feature.

1.2 The Business Value for Your Users

Subscribers pay one fee and replace three separate tool subscriptions. The platform creates compounding retention loops: training makes subscribers better at using the AI tools; AI tool usage drives community engagement; community engagement brings them back to training. Multi-tenancy means you can also license the platform to other operators — turning your SaaS into a platform-of-platforms.

- Single subscription replaces multiple paid tools — high perceived value, low churn
- Claude-powered AI tools make every subscriber measurably better at advertising
- LMS establishes your platform as an authority destination, not just a utility
- Gamification mechanics drive daily active usage and session depth
- Multi-tenancy enables white-labeling and future operator licensing revenue
- Live event calendar creates community urgency and human accountability

1.3 Technical Scope

- Secure multi-role authentication with multi-tenant organization support
- Stripe recurring subscription billing with tenant-level plan management
- HD video LMS: hosted streaming, progress tracking, and course management
- Real-time event calendar with full Google Calendar OAuth bidirectional sync
- Gamification engine: points, streaks, badges, and real-time WebSocket leaderboards
- Claude AI pipelines: ad copy generator, content builder, interactive coaching agent
- Premium UI/UX: dark/light mode, per-tenant theming, micro-interactions
- Admin panel, analytics dashboard, audit logs, notifications
- Multi-tenancy: subdomain routing, tenant isolation, white-label branding
- AWS infrastructure with CI/CD, autoscaling, and 99.9% uptime architecture

1.4 Delivery Model

Development is structured in seven phases. A working, tested deliverable is provided at the end of every phase. You review and approve before the next phase begins. This eliminates end-of-project surprises and keeps your investment protected throughout.

> *ALL FEATURES ARE IN SCOPE FOR THIS ENGAGEMENT. No capabilities have been deferred to post-launch. The platform ships complete.*

1.5 Billing at a Glance

Hourly at $16 USD per hour, invoiced weekly based on accumulated hours with an itemized timesheet. Each phase carries a not-to-exceed (NTE) hour cap — your maximum cost per phase is defined and agreed before work begins.

2. Platform Analysis & Competitive Benchmarking

2.1 AdLevel.ai Deep Analysis

AdLevel.ai is an AI-driven advertising optimization platform. The following capabilities from its model directly shaped the architecture of your platform:

- AI-generated ad copy: headline, body text, and CTA variations at scale
- Multi-platform campaign performance analysis (Meta, Google, TikTok)
- AI-powered budget and targeting recommendations with creative fatigue detection
- KPI dashboard: ROAS, CPM, CTR, CPA, and spend vs. performance visualization
- Subscription tiers gated by usage: API calls, campaigns tracked, team seats
- Workspace management for agency-level team access

2.2 Competitive Landscape

Platform	Core Strength	Notable Gap	What Your Platform Takes From It
AdLevel.ai	AI ad toolkit, campaign analytics	No community, training, or gamification	AI tool architecture, usage-based tier gating
Skool	Gamified community + LMS	No AI tools, no advertising focus	Gamification engine, community loops, course structure
Whop	Digital product monetization, clean UX	Not a training or AI platform	Subscription model, marketplace-style dashboard
Kajabi	Full LMS + email + community	No AI, no ad tools, no multi-tenancy	LMS architecture, progress tracking, event scheduling
Your Platform	All of the above + multi-tenancy	This is the opportunity	The synthesis: AI + LMS + community + gamification + white-label

2.3 Feature Scope — All Features Included

Feature Area	What Is Delivered
AI Tools	3 Claude-powered tools with real-time streaming, saved history, admin-editable prompts, usage tracking
Video LMS	Full course upload, HD streaming via Mux, progress tracking, completion certificates, module management
Leaderboard	Real-time WebSocket live updates, daily/weekly/all-time rankings, animated rank changes

Feature Area	What Is Delivered
Calendar	Full OAuth bidirectional Google Calendar sync, iCal export, RSVP system, event reminders
Analytics	BI dashboard: DAU/MAU, revenue, cohort analysis, course completion rates, AI tool usage metrics
Gamification	Points engine, daily streaks, badge system, level progression, activity feed
Multi-Tenancy	Subdomain routing, per-tenant white-label branding, tenant admin roles, data isolation, tenant billing
Admin Panel	Full user management, course management, AI prompt editor, subscription oversight, broadcast notifications
Notifications	In-app (WebSocket push) + email (Resend), user-configurable preferences
Community Portal	Member directory, activity feed, engagement metrics, community leaderboard

3. Technology Stack

> *Every technology below is production-proven, well-documented, and selected for scalability, developer velocity, and long-term maintainability.*

3.1 Frontend

Layer	Technology	Rationale
Framework	Next.js 14 (App Router)	Native to the Vercel/Linear aesthetic specified; built-in SSR, image optimization, per-tenant routing
Language	TypeScript	Eliminates runtime bugs; essential for a large, maintainable multi-tenant codebase
Styling	Tailwind CSS 3	Utility-first CSS; per-tenant theming via CSS variables; used by Vercel and Linear
Components	ShadCN/UI	Fully customizable Tailwind-native components; dark/light mode and tenant theming built in
State	Zustand	Lightweight, TypeScript-native global state; tenant context managed globally
Data	TanStack Query	Caching, background refetch, pagination; tenant-scoped query keys
Forms	React Hook Form + Zod	Best-in-class form performance; shared validation schema between frontend and backend
Charts	Recharts + Tremor	Recharts for custom data viz; Tremor for BI dashboard components
Video	Mux Player React	Adaptive streaming, thumbnails, and DRM automatically; tenant-level video access control
Calendar	FullCalendar React	Full-featured calendar; Google Calendar OAuth adapter built in
Animations	Framer Motion	Micro-interactions, leaderboard rank animations, page transitions
Auth	Clerk embedded components	OAuth, MFA, social login, organization (tenant) management; styled per tenant
Rich Text	TipTap	Modular editor for course content and AI output editing

3.2 Backend

Layer	Technology	Rationale
Runtime	Node.js 20 LTS	Handles high-concurrency AI streaming and WebSocket connections efficiently
Framework	NestJS	TypeScript-native, modular; multi-tenant middleware and guards integrate cleanly
ORM	Prisma	Schema-first; tenant-scoped queries via row-level security or tenant_id patterns
Queues	BullMQ + Redis	Background processing for AI tasks, emails, video webhooks, gamification events
Validation	Zod (shared)	One schema on both frontend and backend; no duplication or drift
Email	Resend + React Email	Tenant-branded transactional email; React Email templates per tenant
Uploads	AWS S3 (presigned URLs)	Direct browser-to-S3 upload; tenant-prefixed S3 key namespacing
Auth	Clerk backend SDK	Organization-based multi-tenancy built into Clerk; tenant context in every JWT
Payments	Stripe Node SDK	Per-tenant Stripe Connect or shared subscription management
Routing	Subdomain middleware	Resolves tenant from subdomain (tenant.platform.com) on every request

3.3 AI Stack (Claude API)

Component	Choice	Notes
Primary LLM	Anthropic Claude (claude-sonnet-4-20250514)	As specified; best for structured output and instruction-following
Streaming	Server-Sent Events (SSE)	Real-time token streaming to the UI; tenant-scoped usage tracking
Tool Use	Claude tool_use feature	Structured JSON output for ad builder consistency
Prompt Registry	Admin-editable DB templates per tenant	Each tenant can customize AI tool prompts without code changes
Coaching Agent	Redis conversation store	Multi-turn context per user per tenant
Cost Controls	Per-user + per-tenant token logging	Usage tracked and enforced by subscription tier; cost dashboards per tenant

3.4 Multi-Tenancy Architecture

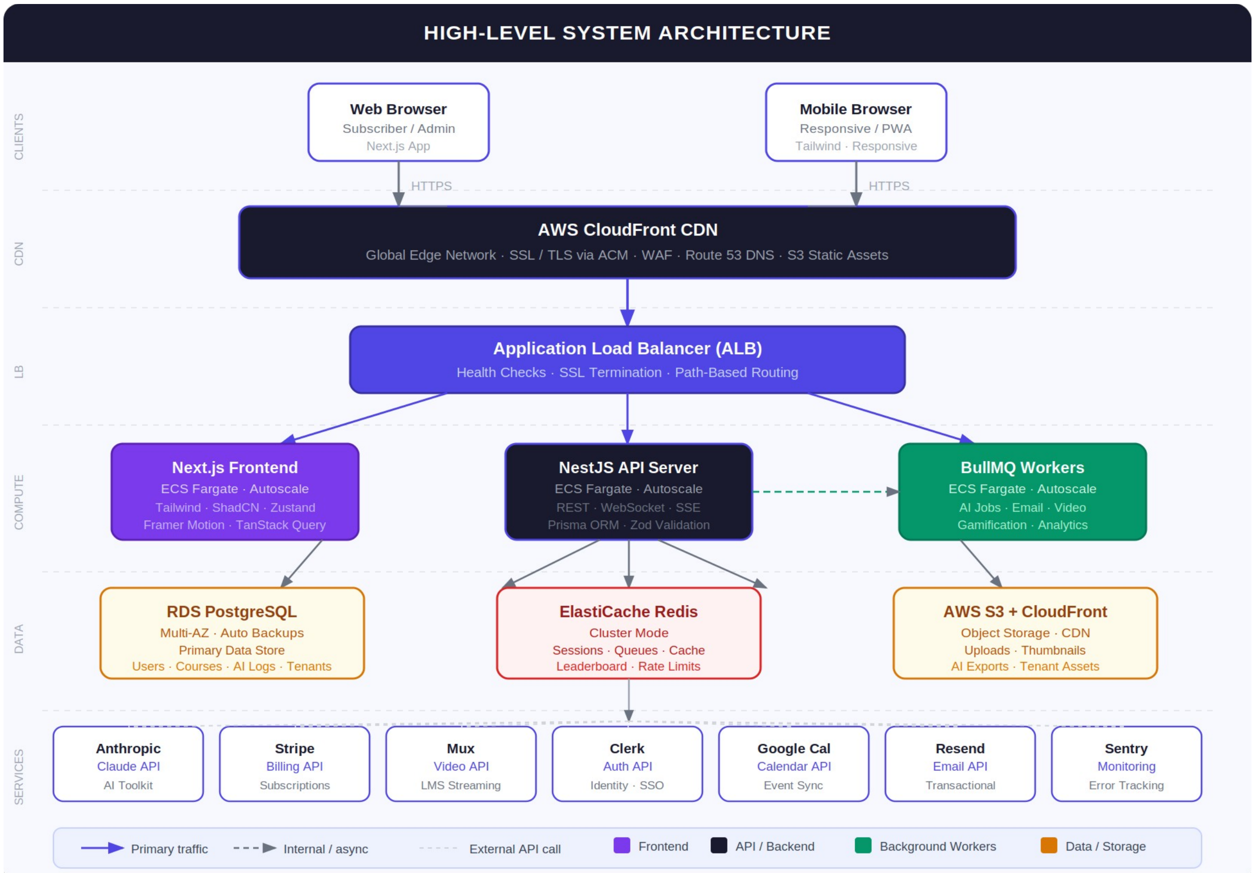
Concern	Approach	Notes
Tenant identification	Subdomain resolution middleware	tenant.platform.com resolves to tenant record on every request
Data isolation	tenant_id foreign key on all core tables	All queries scoped by tenant_id; no cross-tenant data leakage possible
Theming	CSS variables per tenant; stored in DB	Primary color, logo, fonts loaded on tenant resolution; applied via Tailwind
Video access	Mux signed tokens scoped per tenant	Video playback tokens include tenant validation
AI prompts	Tenant-level prompt overrides in ai_prompts table	Tenant admin can customize AI tool behavior
Billing	Tenant-level Stripe subscription	Each tenant has its own Stripe customer and subscription record
Email	Tenant-branded transactional email	From address, logo, and color scheme per tenant
Admin access	SUPER_ADMIN (platform) vs TENANT_ADMIN roles	Platform owner sees all tenants; tenant admin only sees their own

3.5 Infrastructure

Service	Component	Purpose
App Hosting	ECS Fargate	Serverless containers; autoscales per service; zero server management
Database	RDS PostgreSQL Multi-AZ	Managed Postgres; automated backups; automatic failover; tenant data store
Cache	ElastiCache Redis	Sessions, BullMQ queues, leaderboard sorted sets, rate limiting, tenant cache
CDN	CloudFront	Global delivery of assets; tenant-specific cache behaviors via headers
Storage	S3	Tenant-prefixed buckets for uploads, thumbnails, AI exports
SSL	ACM wildcard cert	*.platform.com wildcard covers all tenant subdomains automatically
Secrets	AWS Secrets Manager	All API keys; injected at container runtime; never in code
Monitoring	CloudWatch + Sentry	Infrastructure metrics; application error tracking with tenant context
CI/CD	GitHub Actions + ECR	Automated test, build, deploy pipeline; rolling ECS deploys

4. System Architecture

4.1 High-Level Architecture Diagram



4.2 Multi-Tenant Request Flow

> *MULTI-TENANT REQUEST FLOW User visits tenant1.platform.com | v CloudFront resolves wildcard cert (*.platform.com) | v NestJS TenantMiddleware: extracts subdomain "tenant1" | v DB lookup: SELECT * FROM tenants WHERE subdomain = 'tenant1' | v Tenant context injected into request: tenantId, theme, plan, features | v All downstream queries scoped: WHERE tenant_id = tenantId | v Response rendered with tenant branding (logo, colors, custom domain)*

4.3 AI Request Lifecycle

> *AI REQUEST LIFECYCLE (Ad Copy Generator) User submits: product description, target audience, platform, tenant context | v POST /api/ai/ad-copy — Zod schema validation | v Rate limit check (Redis): user daily limit + tenant monthly cap | v Subscription*

> *check: does this user's plan include this tool? | v Prompt loaded from tenant-specific template (or platform default) | v Messages array built: system prompt + tenant context + user input | v Streaming request to Anthropic Claude API | v SSE streams tokens to browser in real time | v On complete: log usage (tokens, cost, user, tenant) to DB | v Output saved to user's AI history (tenant-scoped)*

4.4 Real-Time Leaderboard Flow

> *REAL-TIME LEADERBOARD FLOW User completes action (lesson, AI tool use, event RSVP, daily login) | v Gamification event fired (event-driven, async via BullMQ) | v Worker calculates points + updates user_points table | v Redis ZADD updates sorted set: leaderboard:{tenant_id}:{period} | v WebSocket gateway broadcasts rank change to all connected clients in tenant | v Frontend receives event: leaderboard row animates to new position (Framer Motion)*

5. Feature Breakdown by Module

MODULE 1: Authentication & User Management

Property	Detail
Purpose	Secure identity, multi-tenant organization management, and role-based access.
Frontend Scope	Login, register, forgot password, profile settings, avatar upload, organization switcher, role-based UI
Backend Scope	Clerk integration, user sync webhooks, roles (SUPER_ADMIN/TENANT_ADMIN/MEMBER/FREE), JWT guards, tenant context middleware
Database	users, user_profiles, tenants, tenant_memberships, roles
External APIs	Clerk API, AWS S3
Complexity	High
Est. Hours	55–70 hrs

MODULE 2: Multi-Tenancy & White-Labeling

Property	Detail
Purpose	Tenant isolation, subdomain routing, per-tenant branding, and operator administration.
Frontend Scope	Tenant theme editor (logo, colors, fonts), custom domain setup UI, tenant onboarding flow, super-admin tenant management panel
Backend Scope	TenantMiddleware (subdomain resolution), tenant CRUD, theme storage and delivery, wildcard SSL handling, data isolation enforcement (tenant_id scoping on all queries)
Database	tenants, tenant_themes, tenant_settings, tenant_features
External APIs	AWS ACM (wildcard cert), Route 53 (subdomain management)
Complexity	Very High
Est. Hours	90–120 hrs

MODULE 3: Subscription & Billing

Property	Detail
Purpose	Monetization layer. Per-tenant subscription management with Stripe.
Frontend Scope	Pricing page, Stripe Checkout flow, billing portal link, upgrade prompts, payment status pages
Backend Scope	Stripe Checkout sessions, webhook handler (subscription lifecycle, payment failure), tenant-level tier enforcement middleware
Database	subscriptions, subscription_tiers, payment_events
External APIs	Stripe API
Complexity	High
Est. Hours	45–60 hrs

MODULE 4: Video LMS

Property	Detail
Purpose	Premium HD video training with full course management and completion tracking.
Frontend Scope	Course library, course detail page, Mux Player, progress bars, module navigation, completion UI, certificates, search and filter
Backend Scope	Course/module/lesson CRUD, Mux upload URL generation, Mux webhook handler, progress tracking API, enrollment management, certificate generation
Database	courses, modules, lessons, enrollments, lesson_progress, course_completions, certificates
External APIs	Mux Video API, Mux webhooks
Complexity	High
Est. Hours	85–115 hrs

MODULE 5: Event Calendar

Property	Detail
Purpose	Full bidirectional Google Calendar sync, event management, and RSVP system.
Frontend Scope	FullCalendar monthly/weekly/agenda views, event creation modal, RSVP flow, Google Calendar OAuth connect, bidirectional sync UI, iCal export
Backend Scope	Event CRUD, Google Calendar OAuth + token refresh, bidirectional sync job (BullMQ), iCal feed (.ics endpoint), RSVP management, reminder scheduling

Property	Detail
Database	events, event_rsvps, calendar_integrations (per user OAuth tokens), calendar_sync_log
External APIs	Google Calendar API (full OAuth), iCal
Complexity	High
Est. Hours	60–80 hrs

MODULE 6: Gamification Engine

Property	Detail
Purpose	Points, streaks, badges, and real-time WebSocket leaderboards driving daily engagement.
Frontend Scope	Live leaderboard (WebSocket), XP/points display, streak counter, badge showcase, activity feed, Framer Motion rank animations, level progress bar
Backend Scope	Event-driven points engine (lesson complete, AI use, RSVP, daily login), streak calculation, badge award logic, Redis ZADD sorted sets for leaderboard, NestJS WebSocket gateway
Database	user_points, point_events, badges, user_badges, streaks, leaderboard_snapshots
External APIs	Internal WebSocket
Complexity	High
Est. Hours	65–85 hrs

MODULE 7: Claude AI Toolkit

Property	Detail
Purpose	Three fully featured Claude-powered tools: Ad Copy Generator, Content Builder, and Coaching Agent.
Frontend Scope	AI tool hub dashboard, per-tool UI with live streaming output, AI output history, saved outputs, token usage meter, multi-turn coaching chat
Backend Scope	Prompt registry (tenant-customizable), Claude API streaming (SSE), rate limiting per user+tenant, usage and cost logging, conversation state (Redis), tenant-scoped AI history
Database	ai_prompts (versioned, tenant-overridable), ai_sessions, ai_messages, ai_usage_logs, ai_outputs
External APIs	Anthropic Claude API
Complexity	Very High

Property	Detail
Est. Hours	85–115 hrs

MODULE 8: Dashboard

Property	Detail
Purpose	Personalized command center: course progress, events, AI usage, leaderboard rank.
Frontend Scope	Welcome widget, course progress, upcoming events, leaderboard rank card, recent AI outputs, quick-launch AI tools, notifications bell, dark/light + tenant theme toggle
Backend Scope	Dashboard aggregation API; notification service; tenant-scoped data
Database	No new tables; aggregates from all modules
External APIs	Internal
Complexity	Medium
Est. Hours	35–48 hrs

MODULE 9: Admin Panel

Property	Detail
Purpose	Full platform management for both super-admin (platform owner) and tenant-admin.
Frontend Scope	User management, course management, AI prompt template editor, subscription overview, platform analytics, broadcast notifications, tenant management (super-admin only)
Backend Scope	Admin-only route guards, bulk operations, metrics aggregation, admin audit logging, tenant CRUD (super-admin), user impersonation (audit-logged)
Database	admin_actions log
External APIs	Stripe API, Mux API
Complexity	High
Est. Hours	60–75 hrs

MODULE 10: Notifications

Property	Detail
Purpose	In-app and email notifications across all tenant contexts.
Frontend Scope	Notification bell with unread badge, notification panel, per-user preference settings
Backend Scope	Notification service (in-app WebSocket + email), BullMQ email queue, tenant-branded email templates
Database	notifications, notification_preferences
External APIs	Resend (tenant-branded email)
Complexity	Medium
Est. Hours	30–42 hrs

MODULE 11: Analytics & Reporting

Property	Detail
Purpose	BI dashboard for operators; personal progress dashboard for subscribers.
Frontend Scope	Admin BI dashboard (DAU/MAU, revenue, cohort analysis, course completion, AI usage, tenant metrics); user personal analytics (progress, points history, AI usage)
Backend Scope	PostgreSQL aggregation queries + materialized views, CSV export, tenant-scoped metrics
Database	analytics_events (partitioned by month), materialized views per tenant
External APIs	Internal
Complexity	High
Est. Hours	50–68 hrs

6. Database Design

6.1 Multi-Tenant Data Model

> *TENANT ISOLATION PATTERN   Every core table includes: tenant_id UUID NOT NULL REFERENCES tenants(id)   Example query (always scoped):   SELECT * FROM courses WHERE tenant_id = $tenantId AND status = 'PUBLISHED'   Row-level security (RLS) enforced at application layer via Prisma middleware.   No query reaches the database without a tenant_id scope.*

6.2 Entity Relationships

> *CORE ENTITY RELATIONSHIPS   tenants (1) ──< users (many)   [via tenant_memberships]   tenants (1) ──< courses (many)   tenants (1) ──< events (many)   tenants (1) ──< ai_prompts (many) [tenant overrides]   tenants (1) ──< subscriptions   users (1) ──< enrollments ──> courses   courses (1) ──< modules ──< lessons   users (1) ──< lesson_progress ──> lessons   users (1) ──< user_points   users (1) ──< user_badges ──> badges   users (1) ──< ai_sessions ──< ai_messages   events (1) ──< event_rsvps ──> users   users (1) ──< notifications   users (1) ──< audit_logs*

6.3 Core Tables

Table	Key Columns	Notes
tenants	id, name, subdomain, custom_domain, status, created_at	Root tenant record; subdomain used for routing
tenant_themes	id, tenant_id, primary_color, logo_url, font_family, dark_mode_default	Per-tenant white-label branding
tenant_settings	id, tenant_id, feature_flags (JSONB), ai_prompt_overrides, max_members	Feature toggles and limits per tenant
users	id, clerk_id, email, created_at	Global user record; tenant membership is separate
tenant_memberships	id, user_id, tenant_id, role, joined_at	Many-to-many; user can be in multiple tenants
subscriptions	id, tenant_id, stripe_customer_id, stripe_sub_id, tier, status, period_end	One per tenant; status: active/past_due/canceled
courses	id, tenant_id, title, slug, thumbnail_url, status, order_index	tenant_id scopes all course queries
modules	id, course_id, title, order_index	Ordered sections within a course
lessons	id, module_id, title, mux_asset_id, mux_playback_id, duration_seconds, order_index, status	Mux IDs populated after upload completes
enrollments	id, user_id, course_id, tenant_id,	completed_at null until all

Table	Key Columns	Notes
	enrolled_at, completed_at	lessons done
lesson_progress	id, user_id, lesson_id, tenant_id, watch_seconds, completed, completed_at	Updated every 30s during playback
events	id, tenant_id, title, start_at, end_at, timezone, google_event_id, max_rsvps	All times UTC; google_event_id for sync
calendar_integrations	id, user_id, tenant_id, google_access_token, refresh_token, expires_at	Per-user Google OAuth tokens; refresh managed by BullMQ job
user_points	id, user_id, tenant_id, points, source_type, source_id, created_at	Append-only audit trail of point events
streaks	id, user_id, tenant_id, current_streak, longest_streak, last_activity_date	Updated on each qualifying action
badges	id, tenant_id, name, icon_url, trigger_type, trigger_value	Tenant-customizable badge definitions
ai_prompts	id, tenant_id (nullable), feature_key, system_prompt, template, model, version	NULL tenant_id = platform default; tenant row overrides
ai_sessions	id, user_id, tenant_id, feature_key, created_at	Groups messages for coaching agent context
ai_usage_logs	id, user_id, tenant_id, feature_key, input_tokens, output_tokens, cost_usd, created_at	Per-user + per-tenant billing and throttling
notifications	id, user_id, tenant_id, type, title, body, read, action_url, created_at	Tenant-scoped in-app notifications
audit_logs	id, user_id, tenant_id, action, entity_type, entity_id, metadata, ip, created_at	Immutable; never updated or deleted
analytics_events	id, user_id, tenant_id, event_type, properties, created_at	Append-only; partitioned by month

7. AWS Infrastructure Design

7.1 Environment Strategy

Environment	Purpose	Infrastructure	Deploy Trigger
local	Development	Docker Compose (Postgres, Redis, Next.js, NestJS)	Manual
staging	QA + preview	Single ECS task per service, RDS t3.micro, Redis t3.micro	Push to main (automated)
production	Live platform	Multi-AZ ECS, RDS Multi-AZ, Redis cluster	Manual approval after staging

7.2 Estimated Monthly Costs

AWS Service	MVP Cost/mo	Growth Cost/mo	Notes
ECS Fargate	~$55	~$220	3 services (frontend, API, worker) × 2 tasks min
RDS PostgreSQL (t3.medium)	~$55	~$200	Multi-AZ doubles cost in production
ElastiCache Redis	~$18	~$70	Single node MVP; cluster at growth
CloudFront + S3	~$15	~$50	Wildcard cert covers all tenant subdomains
ALB (×2)	~$20	~$40	One per service
NAT Gateway	~$35	~$70	Charged per GB; often underestimated
Route 53 + ACM	~$6	~$12	Hosted zone + wildcard cert + subdomain records
Secrets Manager	~$4	~$8	Per secret per month
CloudWatch + ECR	~$12	~$35	Logs, metrics, image storage
TOTAL AWS	**~$220/mo**	**~$705/mo**	Excludes Mux, Clerk, Stripe, Claude API

7.3 Third-Party Service Costs (Client-Owned)

> *These are platform operating costs paid directly by the client to third-party providers. They are not part of development billing.*

Service	Est. Monthly	Notes
Mux Video	\$30–\$100/mo	Scales with video hours stored and delivered
Clerk Auth	\$0–\$25/mo	Free up to 10,000 MAU; Pro plan beyond; Organizations feature included
Anthropic Claude	\$50–\$300/mo	Highly variable with subscriber usage volume
Resend Email	\$0–\$20/mo	Free tier covers early MVP
Stripe	2.9% + \$0.30/transaction	No monthly fee
Sentry	\$0–\$26/mo	Free tier covers MVP monitoring

8. Development Timeline

8.1 Phase Overview

Phase	Weeks	Deliverable	End State
Phase 0: Foundation	1–2	Project setup, AWS skeleton, design system	Dev environment, CI/CD, design tokens, wildcard SSL
Phase 1: Auth, Tenancy & Billing	3–6	Auth + multi-tenancy + Stripe subscriptions	Tenants can register, brand, subscribe, and gate content
Phase 2: Video LMS	7–10	Full video LMS with HD streaming and tracking	Courses playable; progress tracked; certificates issued
Phase 3: AI Toolkit	11–14	3 Claude AI tools with real-time streaming	Ad copy, content builder, and coaching agent fully live
Phase 4: Calendar + Gamification	15–17	Google Calendar sync + real-time leaderboards	RSVP live; leaderboard updates in real time via WebSocket
Phase 5: Admin + Analytics	18–20	Full admin panel and BI analytics dashboard	Operator manages all users, courses, tenants, and metrics
Phase 6: Polish + Launch	21–26	QA, security, performance, production deploy	Platform live in production; monitoring active; launch-ready

8.2 Week-by-Week Breakdown

Wk	Phase	Deliverables
1	Foundation	Monorepo structure, Docker Compose, GitHub repo, ESLint/Prettier/TypeScript, CI skeleton
2	Foundation	AWS VPC, RDS, Redis, ECS skeleton, wildcard ACM cert, Tailwind + ShadCN + dark mode, design tokens
3	Auth+Tenancy	Clerk integration, user registration/login, JWT guard, role middleware, tenant middleware (subdomain resolution)
4	Auth+Tenancy	Tenant CRUD, theme engine (CSS variables per tenant), tenant onboarding flow, white-label UI
5	Auth+Tenancy	Stripe customer creation, Checkout session, pricing page, subscription status sync, tier middleware
6	Auth+Tenancy	Stripe webhooks (subscription lifecycle), billing portal, payment failure handling, tenant billing isolation
7	LMS	Course/module/lesson Prisma models, admin CRUD UI, course listing page, tenant-scoped queries
8	LMS	Mux integration, presigned upload flow, Mux webhook handler (asset.ready), Mux Player component

Wk	Phase	Deliverables
9	LMS	Progress tracking API (30s heartbeat), lesson completion logic, enrollment, certificate generation
10	LMS	Course detail UX, module navigation, completion indicators, search and filter, mobile responsiveness
11	AI Toolkit	Prompt registry (tenant-overridable), Claude SDK streaming, SSE endpoint, rate limiter, usage logging
12	AI Toolkit	Ad copy generator: UI, streaming output, prompt engineering, history saving, token meter
13	AI Toolkit	Content builder tool (posts, emails, scripts), multi-format output, saved outputs
14	AI Toolkit	Coaching agent: multi-turn chat UI, Redis session store, tenant coaching persona, context management
15	Calendar	FullCalendar integration, event CRUD (tenant-scoped), RSVP system, iCal export
16	Calendar	Google Calendar full OAuth: connect flow, token refresh job (BullMQ), bidirectional sync, event reminders
17	Gamification	Points engine, streak logic, badge system, Redis ZADD leaderboard, NestJS WebSocket gateway, Framer Motion animations
18	Admin	User management, course management, AI prompt template editor, tenant management (super-admin)
19	Admin	BI analytics dashboard (Recharts/Tremor), notification system (in-app + Resend email)
20	Admin	Broadcast notifications, subscription oversight, admin audit logs, user impersonation (audit-logged)
21	Polish	Unified dashboard, onboarding flow, empty states, micro-interactions, mobile audit
22	QA	Full regression testing, cross-browser, tenant isolation verification, WebSocket load testing
23	QA	Edge case testing, Stripe webhook reliability, Google Calendar sync edge cases, Mux failure handling
24	Security	Security audit: RBAC review, SQL injection, rate limits, tenant data isolation verification
25	Security	Performance: query optimization (EXPLAIN ANALYZE), Lighthouse audit, load testing (k6)
26	Launch	Production DNS + wildcard subdomains, CloudFront deploy, ECS production rollout, Sentry + CloudWatch setup, go-live

9. Project Gantt Chart

Phase	Window	Focus	Weeks 1 — 26
Phase 0: Foundation	W1–W2	AWS, design system, CI/CD	
Phase 1: Auth, Tenancy & Billing	W3–W6	Clerk, multi-tenancy, Stripe	
Phase 2: Video LMS	W7–W10	Mux, courses, progress, certs	
Phase 3: AI Toolkit	W11–W14	Claude streaming, 3 AI tools	
Phase 4: Calendar + Gamification	W15–W17	Google Cal OAuth, leaderboard	
Phase 5: Admin + Analytics	W18–W20	Admin panel, BI dashboard	
Phase 6: Polish + Launch	W21–W26	QA, security, production	

MILESTONES: M1 W6 — Auth + Tenancy + Billing live | M2 W10 — LMS complete | M3 W14 — AI tools live | M4 W17 — Calendar + Gamification | M5 W20 — Admin + Analytics | M6 W26 — Production launch

10. Development Hour Estimation

10.1 Estimates by Module

Module / Phase	Conservative	Realistic	Maximum
Phase 0: Foundation + Infrastructure	55	70	85
Auth, User Management & Multi-Tenancy	100	130	165
Subscriptions & Billing (Stripe)	45	58	72
Video LMS (Mux + Certificates)	80	108	138
Event Calendar (Full Google Cal OAuth)	60	78	98
Gamification + Real-Time Leaderboard	60	80	102
Claude AI Toolkit (3 full tools)	80	108	138
Dashboard	32	44	56
Admin Panel (Super + Tenant Admin)	62	78	96
Notifications	28	40	52
Analytics & BI Dashboard	48	64	82
QA + Testing (integrated throughout)	60	82	108
Security Hardening	25	36	48
DevOps + CI/CD + Deployment	38	52	68
Buffer (revisions, scope, bugs)	55	82	125
TOTAL	**828**	**1,110**	**1,433**

> *NTE CAPS ARE SET CONSERVATIVELY. Work will be invoiced at actual hours accumulated weekly. If a module comes in under its NTE cap, you save the difference. The NTE caps protect your maximum exposure per phase.*

11. Billing Model & Payment Plan

11.1 Rate

> *Hourly Rate: $16 USD per hour | Billing: Weekly accumulated hours | Invoice: Itemized timesheet every Friday*

This rate is positioned as competitive Philippine-based talent, delivering senior-level full-stack engineering with specializations in AI integration (Anthropic Claude), multi-tenant SaaS architecture, AWS cloud infrastructure, and DevOps. It represents approximately 78–82% cost savings compared to equivalent US or Australian developer rates ($75–$95/hr) for the same technical scope and quality standard.

11.2 Phase-Level Not-to-Exceed (NTE) Caps

Each phase has a fixed NTE cap. You know your maximum cost before any phase begins. If work is completed under the cap, you are billed only for actual hours. Scope changes are scoped and approved separately before proceeding.

Phase	Scope	NTE Hours	Max Cost @ $16/hr	When Billed
Phase 0	Foundation, AWS, Design System	80 hrs	$1,280	Weekly during phase
Phase 1	Auth + Multi-Tenancy + Billing	170 hrs	$2,720	Weekly during phase
Phase 2	Video LMS	125 hrs	$2,000	Weekly during phase
Phase 3	Claude AI Toolkit (3 tools)	130 hrs	$2,080	Weekly during phase
Phase 4	Calendar + Gamification	120 hrs	$1,920	Weekly during phase
Phase 5	Admin Panel + Analytics	120 hrs	$1,920	Weekly during phase
Phase 6	QA, Security, Production Launch	155 hrs	$2,480	Weekly during phase
TOTAL NTE	**Full Platform**	**900 hrs**	**$14,400**	**Weekly throughout**

> *COST COMPARISON: An equivalent developer in the US or Australia would charge $67,500–$85,500 for the same scope (900 hrs × $75–$95/hr). This engagement delivers the same technical quality at $14,400 maximum — a saving of $53,000–$71,000.*

11.3 Weekly Billing Process

1. Hours logged daily with task-level descriptions (e.g., '3.5 hrs — Tenant middleware + subdomain resolution')

2. Timesheet submitted every Friday with accumulated hours for that week
3. Invoice issued same day as timesheet; itemized by task and module
4. Payment due within 5 business days of invoice
5. Work continues into the next week without interruption provided payments are current

11.4 Kickoff Deposit

A deposit of $640 USD (equivalent to 40 hours at $16/hr) is required before work begins. This deposit is applied against the first two weeks of billing and confirms project commitment from both parties.

11.5 Scope Change Policy

Any requirement not included in this proposal is treated as a change request. Change requests are scoped, estimated in writing, and require written approval before work begins. They are billed separately at $16/hr and do not affect existing phase NTE caps.

11.6 Payment Methods Accepted

- Upwork (if contracted via platform)
- Wise (international bank transfer — recommended for lowest fees)
- PayPal
- Direct bank transfer (USD or PHP)

12. Risk Assessment & Mitigation

Risk	Likelihood	Impact	Mitigation
Scope additions mid-build	High	High	Written change request process; written approval and separate NTE before any out-of-scope work begins
Claude API cost overrun at scale	Medium	High	Per-user + per-tenant daily token limits; usage logged from day one; tier-gated access; cost dashboard
Multi-tenancy data isolation failure	Low	Very High	All queries require tenant_id; Prisma middleware enforces this; automated tenant isolation tests in CI
Google Calendar OAuth token expiry	Medium	Medium	BullMQ refresh job runs 30 min before expiry; graceful re-auth UI flow if refresh fails
Mux video webhook failure	Low	High	Retry logic with backoff; polling fallback; admin alert on stuck assets
Stripe webhook idempotency failure	Low	High	Idempotency keys on all handlers; event deduplication in DB
Wildcard SSL subdomain propagation delay	Low	Medium	ACM wildcard cert eliminates per-tenant cert provisioning; Route 53 automation handles subdomain records
Performance degradation with many tenants	Low	High	tenant_id indexes on all tables from day one; materialized views per tenant; Redis tenant context caching
Delayed client feedback slowing timeline	Medium	Medium	Weekly deliverables per phase; defined 3-day review window; async feedback via Loom
Credential or API key exposure	Low	Very High	AWS Secrets Manager for all keys; no secrets in code; regular dependency security audits

13. Security Architecture

13.1 Multi-Tenant Security

- All database queries include tenant_id scoping enforced by Prisma middleware — no query executes without a tenant context
- Cross-tenant data access is architecturally impossible at the query layer; additionally tested in CI with automated isolation tests
- Tenant admin cannot access super-admin routes; RBAC enforced at route level with NestJS guards
- Mux video playback tokens are scoped per tenant; a token from one tenant cannot play another tenant's content

13.2 Authentication Security

- Clerk handles all auth complexity: password hashing, session management, OAuth 2.0, brute-force protection, MFA, and Organization management
- JWTs are short-lived (1 hour); Clerk manages refresh tokens automatically; tenant context embedded in JWT claims
- Session revocation supported instantly via Clerk dashboard for compromised accounts

13.3 API Security

- All routes protected by AuthGuard + TenantGuard by default; public routes explicitly decorated
- Role-based access control (RBAC) enforced at route level with custom NestJS guards
- All API input validated by Zod schemas; malformed requests rejected before business logic
- Rate limiting: global (100 req/min per IP) + per-user AI limits by subscription tier + per-tenant monthly caps
- CORS: production whitelist only; Helmet.js security headers on all responses

13.4 Infrastructure Security

- VPC: all ECS tasks and databases in private subnets; only ALB is publicly accessible
- IAM: ECS task roles with least-privilege policies; no hardcoded credentials anywhere
- AWS Secrets Manager: all API keys injected at container runtime
- S3: all buckets private; content delivered via CloudFront with signed URLs
- RDS: automated daily backups, 7-day retention (30-day production); point-in-time recovery enabled

13.5 AI-Specific Security

- Prompt injection protection: system prompts wrap user input with explicit boundary markers
- AI output sanitized with DOMPurify before rendering in the browser
- No subscriber PII injected into AI prompts; tenant data handling disclosed in Terms of Service

13.6 Privacy Compliance

- Privacy policy and Terms of Service required before launch (Termly.io recommended for multi-tenant compliance)
- Data export endpoint for GDPR Article 20 compliance; tenant-scoped export
- Account and tenant deletion: soft delete with 30-day grace period; full hard delete on verified request

14. Deployment & CI/CD Strategy

14.1 Pipeline Overview

> *GITHUB ACTIONS CI/CD PIPELINE On pull_request → staging: ├── ESLint + Prettier check ├── TypeScript type check ├── Unit tests (Jest) ├── Tenant isolation tests └── Build verification (Next.js + NestJS) On merge to staging: ├── All checks above ├── Build Docker images (frontend + API + worker) ├── Push to AWS ECR └── Rolling deploy to ECS staging On manual approval → main (production): ├── All checks above ├── RDS automated snapshot (pre-deploy backup) ├── Prisma migrations run ├── Rolling deploy to ECS production (zero-downtime) └── Slack/email notification on success or failure*

14.2 Rollback Strategy

- ECS retains the previous task definition; rollback is a single command redeploying the last known-good image (under 3 minutes)
- Database migrations are backwards-compatible by design; no destructive changes in the same deploy as the code that depends on them
- Pre-deploy RDS snapshot ensures a clean restore point exists before every production deployment

14.3 Monitoring & Uptime

- Sentry: frontend and backend error tracking; alerts on new error classes with tenant context
- CloudWatch: ECS health, RDS performance insights, Redis metrics, custom alarms
- Health endpoint (/api/health): DB, Redis, and external API status
- External uptime monitoring (Uptime Robot): SMS and email alerts on downtime

15. Post-Launch Maintenance & Support

15.1 Immediate Post-Launch (Weeks 1–4)

- Daily monitoring of Sentry, CloudWatch, and Stripe dashboard
- Critical bugs resolved within 24 hours; major bugs within 72 hours
- Claude API cost monitoring per tenant against projections; rate limits adjusted if needed
- Tenant isolation verification: confirm no cross-tenant data access in production
- Query performance review: optimize slow queries from real production load

15.2 Ongoing Monthly

- npm dependency updates and security patches; applied in staging before production
- Database health: vacuum analyze, index health, backup verification
- AI prompt review: iterate based on user behavior and tenant feedback
- Monthly cost audit: Claude API, Mux, Clerk, AWS per tenant

15.3 Optional Retainer Plans

Plan	Monthly Rate	Coverage
Basic	$800–$1,200/mo	Monitoring, critical bug fixes, monthly dependency updates, infrastructure health
Standard	$2,000–$3,000/mo	Basic + up to 20 hrs/month feature development
Premium	$3,500–$5,000/mo	Standard + priority SLA + up to 40 hrs/month feature and growth work

16. Strategic Recommendations

16.1 Recommended Build Sequence

Build in this exact order. Each phase unlocks value for the next.

6. Auth + Multi-Tenancy + Billing first. The entire platform depends on tenants being able to register and subscribe. Nothing else has value without this foundation.
7. LMS second. This is the first paid value delivery. Subscribers need content before everything else.
8. AI Toolkit third. This is the differentiated moment that drives word of mouth. Get the ad copy generator live as soon as the LMS is working.
9. Event Calendar fourth. Live sessions prove the community is growing and active.
10. Gamification fifth. Once content and AI tools exist, gamification creates compounding retention on top of them.
11. Admin Panel and Analytics trail the user-facing features by one to two weeks without affecting the subscriber experience.

16.2 Why Multi-Tenancy Now, Not Later

Retrofitting multi-tenancy onto an existing single-tenant codebase is one of the most expensive and risky refactors in SaaS engineering. Every table needs tenant_id columns, every query needs scoping, every auth flow needs organizational context, and every API endpoint needs tenant guards. Building it from the start costs 90–120 hours. Retrofitting it later costs 3–5x that — in addition to a period of downtime risk and thorough re-testing. The decision to include it in the MVP is the correct one.

16.3 Investment Summary

Item	Detail
Hourly Rate	$16 USD / hour
Billing Cycle	Weekly, accumulated hours with itemized timesheet
Kickoff Deposit	$640 USD (applied against first two weeks of billing)
Total NTE — Full Platform	$14,400 USD (900 hours × $16/hr)
Estimated Timeline	26 weeks from kickoff to production launch
MVP Cost (Phases 0–3)	$8,080 USD (NTE 505 hrs × $16/hr)
Scope Change Policy	Written approval required; billed at $16/hr; does not affect phase NTE caps
Post-Launch Support	Optional retainer plans from $800/mo

> *TOTAL SAVINGS VS. US/AU RATES: This engagement delivers the equivalent of $67,500–$85,500 of senior full-stack + AWS + AI + multi-tenant SaaS engineering at a maximum of $14,400 — a saving of $53,000–$71,000 without compromising technical quality, scope, or deliverable*

> *completeness.*

END OF PROPOSAL